

Parallel Practice: Quadratic Functions

Directions: Answer all questions. Unless a question says “Select ALL that apply,” select one answer. The final question requires a written explanation.

1) Select ALL forms of a quadratic equation that match the graph described.

Graph description: The parabola has vertex at $(-2, 5)$, opens upward, has axis of symmetry $x = -2$, and passes through $(0, 9)$.

- A. $f(x) = x^2 + 4x + 9$
- B. $f(x) = (x + 2)^2 + 5$
- C. $f(x) = (x - 2)^2 + 5$
- D. $f(x) = x^2 - 4x + 9$

This question assesses translating graphical features (vertex, axis, opening, and a point) into equation structure.

Rationales:

- **A: Correct.** $(x + 2)^2 + 5 = x^2 + 4x + 9$, matches vertex $(-2, 5)$ and point $(0, 9)$.
- **B: Correct.** Vertex form with $h = -2, k = 5$; $f(0) = 4 + 5 = 9$.
- **C: Incorrect.** Vertex would be $(2, 5)$, not $(-2, 5)$.
- **D: Incorrect.** Axis would be $x = 2$ (from $b = -4 \Rightarrow -\frac{b}{2a} = 2$).

2) Factor implications from end behavior and a known root.

A parabola satisfies: as $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and it has a root at $x = -3$. What can you conclude about the factors of this quadratic?

- A. The factor with a root of $x = -3$ has a negative slope.
- B. The y -intercept is positive.
- C. The other root is positive.
- D. The product of the slopes of the two linear factors is positive.

This question assesses the relationship between end behavior (sign of a) and slopes of linear factors in factored form.

Rationales:

- **A: Incorrect.** The sign of that factor's slope cannot be determined from the given info.
- **B: Incorrect.** The y -intercept depends on c , which is unknown.
- **C: Incorrect.** No information about the second root's sign.
- **D: Correct.** For $f(x) = (mx + p)(nx + q)$, upward opening means $a = mn > 0$, so the factor slopes' product is positive.

3) Select ALL valid factorizations for roots $x = 3$ and $x = -\frac{2}{5}$.

- A. $f(x) = (x - 3)\left(x + \frac{2}{5}\right)$
- B. $f(x) = (5x + 2)(x - 3)$
- C. $f(x) = -(x + 3)(5x - 2)$
- D. $f(x) = (x + 3)\left(x + \frac{2}{5}\right)$

This question assesses understanding that multiplying by nonzero constants preserves roots, and how factor signs encode root signs.

Rationales:

- **A: Correct.** Factors directly encode roots 3 and $-\frac{2}{5}$.
- **B: Correct.** Same roots; scaled by 5 in one factor.
- **C: Incorrect.** Roots would be -3 and $\frac{2}{5}$.
- **D: Incorrect.** Roots would be -3 and $-\frac{2}{5}$.

4) Leading coefficient from a value table.

What is the leading coefficient a of the quadratic?

x	$f(x)$
4	7
5	4
6	-3
7	-14

- A. 2
- B. -2
- C. -1
- D. -4

This question assesses recognizing that the constant second difference equals $2a$.

Work: First differences: $4 - 7 = -3$, $-3 - 4 = -7$, $-14 - (-3) = -11$.

Second differences: $-7 - (-3) = -4$ and $-11 - (-7) = -4$. Thus $2a = -4 \Rightarrow a = -2$.

Rationales:

- **A: Incorrect.**
- **B: Correct.**
- **C: Incorrect.**
- **D: Incorrect.** That would imply $2a = -8$.

5) Convert factored to standard form.

Which standard-form equation is equivalent to $f(x) = (3x + 4)(x - 5)$?

- A. $f(x) = 3x^2 - 11x - 20$
- B. $f(x) = 3x^2 + 11x - 20$
- C. $f(x) = -3x^2 - 11x - 20$
- D. $f(x) = 3x^2 - 11x + 20$

This question assesses multiplying binomials and arranging in standard form.

Work: $(3x + 4)(x - 5) = 3x^2 - 15x + 4x - 20 = 3x^2 - 11x - 20$.

Rationales:

- **A: Correct.**
- **B: Incorrect.** Wrong sign for b .
- **C: Incorrect.** Wrong sign for a .
- **D: Incorrect.** Wrong sign for c .

6) Written reasoning: pattern of change in quadratics.

Even though quadratics include x^2 and bx terms, their pattern of change is neither exponential nor multiplicative.

Prompt. What is the pattern of change in a quadratic function, and why? Use the standard form $f(x) = ax^2 + bx + c$ to support your response.

This question assesses written reasoning about constant second differences and contrast with exponential growth.

Expected elements in a complete response:

- State that for equally spaced x -values, a quadratic exhibits a constant second difference, not a constant ratio.
- Justify algebraically: the first difference of $ax^2 + bx + c$ is linear in x , so the second difference is constant and equals $2a$.
- Contrast with exponentials $g(x) = P \cdot r^x$, which have a constant multiplicative ratio between successive outputs.